

DEA-C01 One-Page Cheat Sheet

Exam: 65 Q (50 scored) · 130 min · pass **720/1000** · Domains: Ingest/Transform **34%**, Store **26%**, Ops **22%**, Security **18%**. Read for the *constraint* (cost / latency / durability / least-privilege / min-ops) → pick the service.

Service comparisons — decide in one line

Pick between	Use the first when...	Use the second when...
Glue ↔ EMR	serverless ETL, low ops	need full Spark/Hive/Presto cluster control
Athena ↔ Redshift	ad-hoc SQL on S3, pay-per-scan	fast repeated BI on loaded data (MPP)
RDS/Aurora ↔ DynamoDB	relational, joins, SQL/OLTP	key-value, massive scale, single-digit-ms
Kinesis Streams ↔ Firehose	custom real-time consumers	no-code delivery to S3/Redshift/OpenSearch
Kinesis ↔ MSK	AWS-native streaming	Kafka ecosystem / portability
Step Functions ↔ MWAA	native, low-ops state machine	complex code-first Airflow DAGs
SQS ↔ SNS ↔ EventBridge	point-to-point queue	pub/sub fan-out event routing + filter + schedule
Glue Catalog ↔ Lake Formation	metadata store	fine-grained row/column permissions
CloudWatch ↔ CloudTrail	metrics/logs/alarms (performance)	API audit (who did what)
DMS ↔ DataSync	DB migration + CDC	file/object transfer
CSV/JSON ↔ Parquet/Avro	interchange / human-readable	columnar analytics (Parquet) row-stream + schema-evolution (Avro)
Provisioned ↔ Serverless	steady heavy load, lowest unit cost	spiky/unknown load, no capacity mgmt

Reflexes (common right answers)

- Cheaper Athena → **Parquet + partitioning + compression** (cost = data scanned).
- Service needs S3 access → **IAM role**, never access keys.
- On-prem DB → AWS, minimal downtime → **DMS full load + CDC**.
- Incremental Glue runs → **job bookmarks**. Slow Spark → **skew / small files / wrong join**.
- Encrypt everything at rest → **SSE-KMS**; secrets → **Secrets Manager**.
- Fine-grained lake access (row/column) → **Lake Formation**.
- Decouple + don't lose work → **SQS + DLQ**; fan-out → **SNS**; route events → **EventBridge**.
- History in a dimension → **SCD Type 2**. Safe re-runs → **idempotency** (stable natural key).
- HA → **Multi-AZ**; DR → **multi-region**.

The project (build a slice each day)

CSV/JSON → S3 raw → Glue Crawler+Catalog → Glue ETL (clean/dedup/join → Parquet)
→ S3 curated → Athena / Redshift → Step Functions + CloudWatch

✔ partition year/month/day · ✔ KMS on every zone · ✔ scoped IAM role · ✔ retry + DLQ · ✔ data-quality check · ✔ lifecycle policy · ✔ Lake Formation column perms · ✔ note the streaming variant

15-day map

1 Cloud+IAM · **2** SQL · **3** Python+formats · **4** DE foundations (star/SCD) · **5** S3 lake · **6** Databases · **7** Glue+Athena · **8** Batch/Spark · **9** Streaming · **10** Orchestration · **11** Migration · **12** Ops+monitoring · **13** Security+governance · **14** Mock+repair (aim 75-80%) · **15** Final mock+review

Every day: **read** → **build** → **drill questions** → **review**. Serverless/on-demand + budget alert; stop provisioned resources after each session.